



Using ADX to Trade Breakouts, Pullbacks, and Reversal Patterns

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What is ADX? (with DMI)

ADX (Average Directional Index)
is an indicator that measures
trend strength shows trend
direction.

ADX tells us whether the **bulls** or
the **bears** are in control.

Benefits of Using ADX

- * Free indicator, no system to buy
- * Quantifies trend strength
- * Allows us to see the strength of **bulls** and **bears** at the same time
- * Tells when the trend is strong enough to trade
- * Tells us when the trend is weak (stand aside)
- * Works on all timeframes and products
- * Can be combined with other indicators



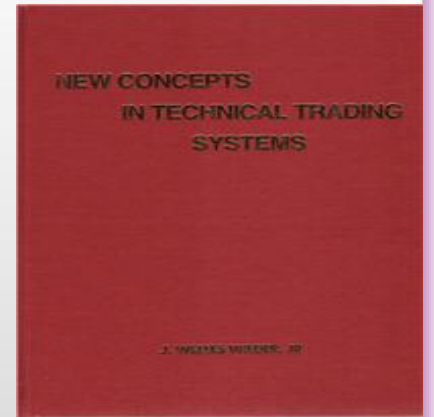
Myth: To make money in the stock market, you must know what the market is going to do.

Truth: You must know what **you're** going to do before the market does what it does.

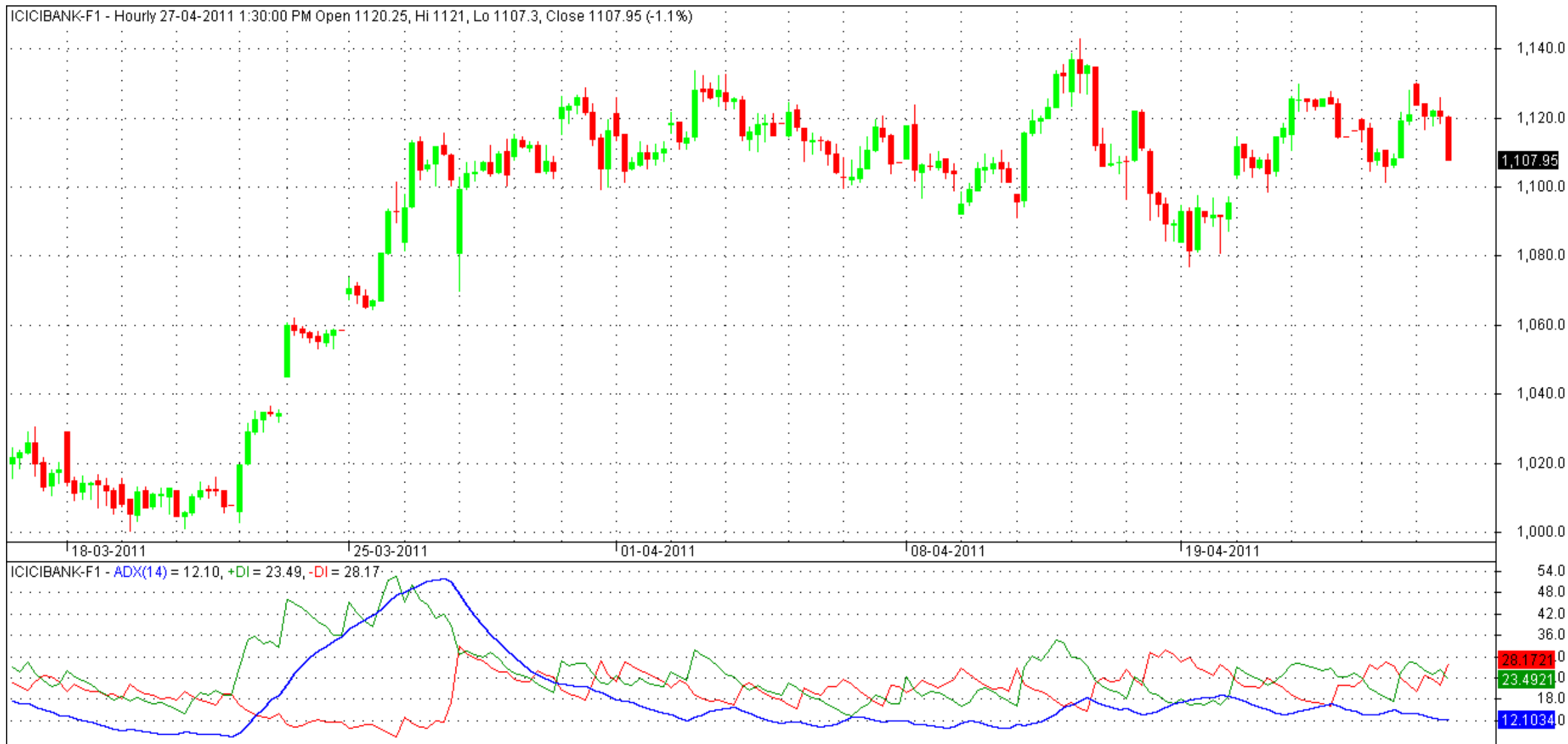
ADX and DMI

Average Directional Movement Index (ADX) was developed by J. Welles Wilder and presented in his book, *New Concepts in Technical Trading Systems* (1978).

ADX is derived from two indicators (developed by Wilder) known as the Positive Directional Movement Indicator (+DMI) and the Negative Directional Movement Indicator (-DMI)



Settings: DMI 14, ADX 14



Directional Movement (DM)

DM is the largest part of today's range that is outside of yesterday's range.

When the largest part of today's range is above yesterday's range, we get positive DM (+DM).

When the largest part of today's range is below yesterday's range, we get negative DM (-DM).

Directional Movement (DM)



+DM



-DM

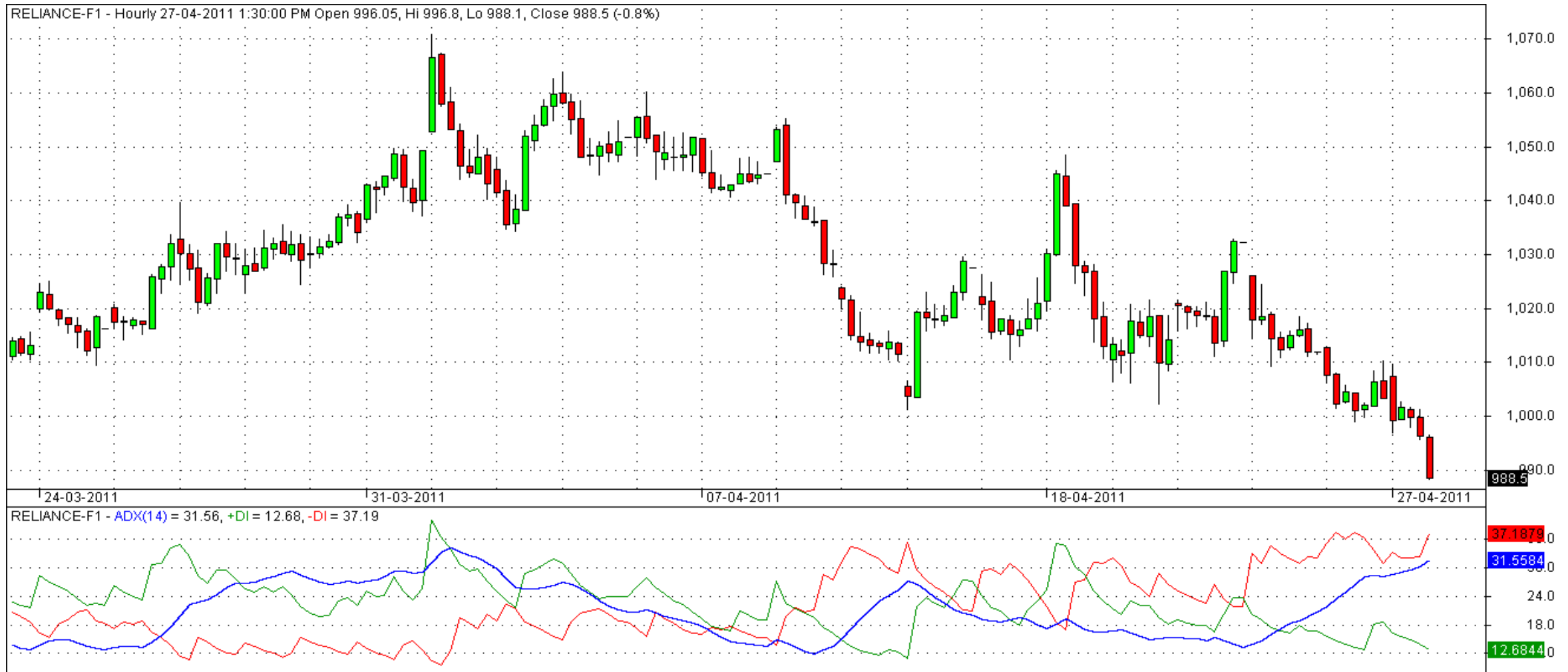


+DM and -DM



No DM

When DMI lines separate—ADX rises, trend strengthening
When DMI lines come together—ADX falls, trend weakening



ADXplained

The primary use of ADX is to measure trend strength.

DMI CONFIRMS trend direction and CONFIRMS price entry/exit signals.

When $ADX > 20$

If +DMI is above -DMI, price is trending up.

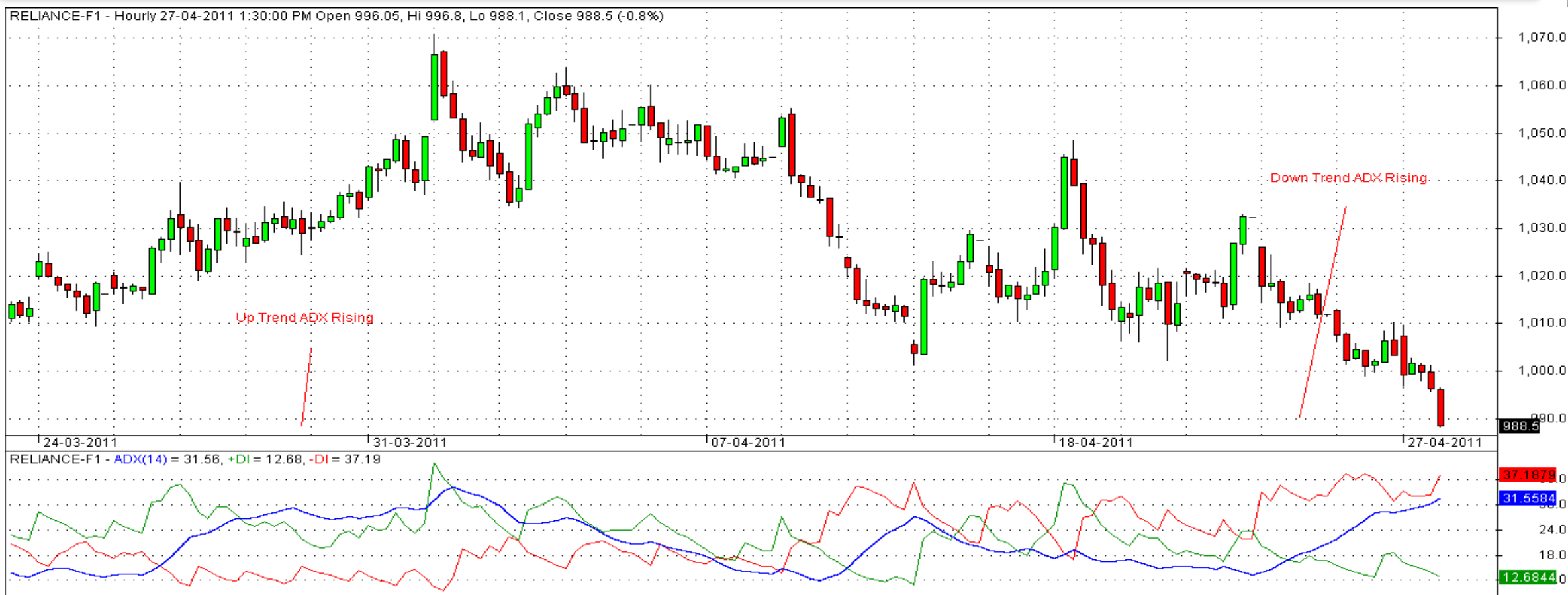
If -DMI is above +DMI, price is trending down.

ADX is No directional

ADX measures the strength of a trend, but doesn't distinguish between uptrends and downtrends.

When there is a strong uptrend, ADX rises.

When there is a strong downtrend, ADX rises.



Short Mode

$ADX > 20$

-DMI above +DMI



Scalp Mode

$ADX < 20$

DMI spikes above 20



Long Mode

$ADX > 20$

+DMI above -DMI



Breakouts
Point #1
Low ADX Periods—No Trend Trading

ADX Trend Strength Rule

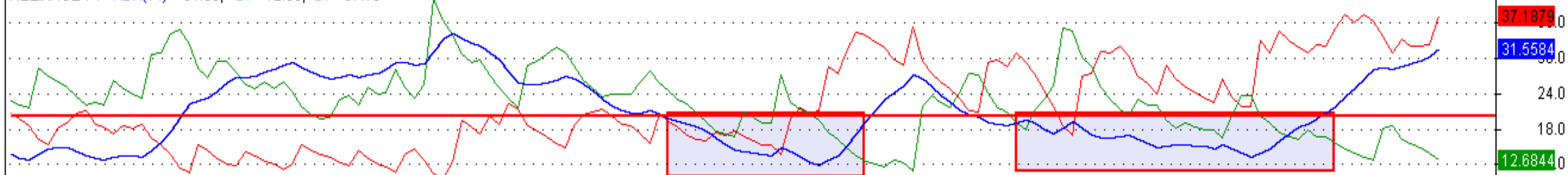
When ADX is above 20, trend strength is strong enough for trend trading strategies

When ADX is below 20, avoid trend trading strategies (optional: scalp trades)

RELIANCE-F1 - Hourly 27-04-2011 1:30:00 PM Open 996.05, Hi 996.8, Lo 988.1, Close 988.5 (-0.8%)



RELIANCE-F1 - ADX(14) = 31.56, +DI = 12.68, -DI = 37.19



Power Grid, you can easily see choppy zones and trend zones



Breakouts
Point # 2
20 Horse Power Minimum

The magic ADX number for a trend to be designated “strong” is 20.

When ADX falls below 20, price is usually in a consolidation period and trend trading strategies will normally fail.

Once ADX rises above 20, the trader can use trend trading strategies.

When ADX is Below both DMI's try finding out patterns on the chart

Price and ADX both generally breakout the pattern together.

ONGC Chart, ADX Below 20 and both DMI's, Price breakout the triangle and later ADX confirmed the trend



BankNifty From 04/03/2011 to 23/03/2011 was trading in channel
On 24th morning we got a breakout with rising ADX



Breakouts

Point # 3

Best Trades Begin From Low ADX Periods

ADX Trendline Rule

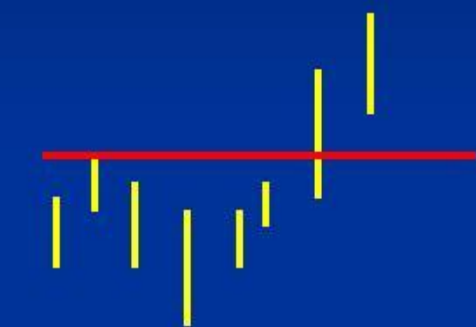
When ADX is below 20 for an extended period,
draw trendlines on price and wait for a breakout

DMI Breakout Rule

A valid breakout up requires a new high in price and
a new high in +DMI (and $+DMI > 20$)

A valid breakout down requires a new low in price
and a new high in -DMI (and $-DMI > 20$)

Key DMI Momentum Signals



+DMI



-DMI

BANKNIFTY-F1 - Hourly 27-04-2011 1:30:00 PM Open 11824.5, Hi 11829, Lo 11768, Close 11771.5 (-0.4%)



Bank Nifty, ADX below both DMI's and price is under range, price and ADX both confirmed the breakouts and Price and +DMI started making higher highs



Bank Nifty, after the earlier trend shown in previous slide, price again entered expanding triangle (pattern), ADX went below both DMI's, confirms the choppy zone.



Bank Nifty, Last Thursday 28th, finally –DMI made higher high and ADX started rising from 20 confirming the down trend, finally price stopped at previous support

ADX Breakout Setup

Lets combine the things which we have learned so far, this is how the setup will look like.

FOR **BUYS** (**SELLS** ARE REVERSED)

1. A 14-period ADX must initially be below 20 for atleast 10 or more bars. This will identify choppy zone.
2. Look for some patterns like rectangle, triangle, channels and etc.
3. Draw trendlines or patterns, and wait for price to break through
4. Before buying wait for ADX to raise above 20
5. Buy once the ADX rises above 20, put stop below the trendline on the price

4. Once filled, enter a protective sell stop below the trendline . Trail the stop as profits accrue and look to exit at the most recent swing high. If you think the market may continue its move, you might exit part of the position at the most recent swing high and tighten stops on the balance.



SILVER-1M - Daily 18-02-2011 Open 47325, Hi 48919, Lo 47325, Close 48462 (2.7%) EMA(Close,20) = 45,492.37

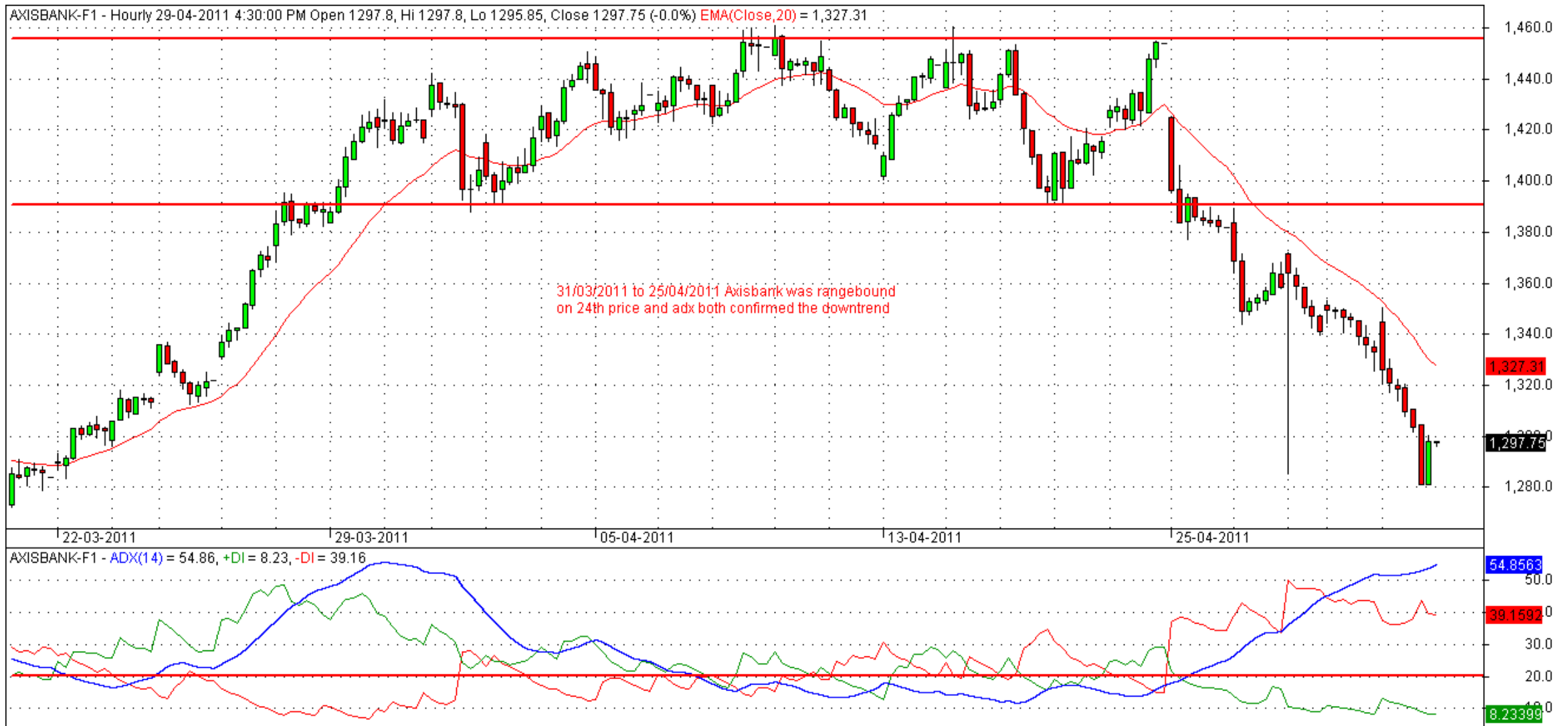


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Silver, 18/02/2011 Price and +DMI made new high, ADX Confirmed the bull trend



ABIRLANUVO, 24th march 2011, price and +DMI together breakout the pattern, later ADX confirmed the Trend.



31/03/2011 to 24/04/2011 Axisbank was range bound on 24th price and ADX both confirmed the downtrend

ADX PullBacks

Point # 4

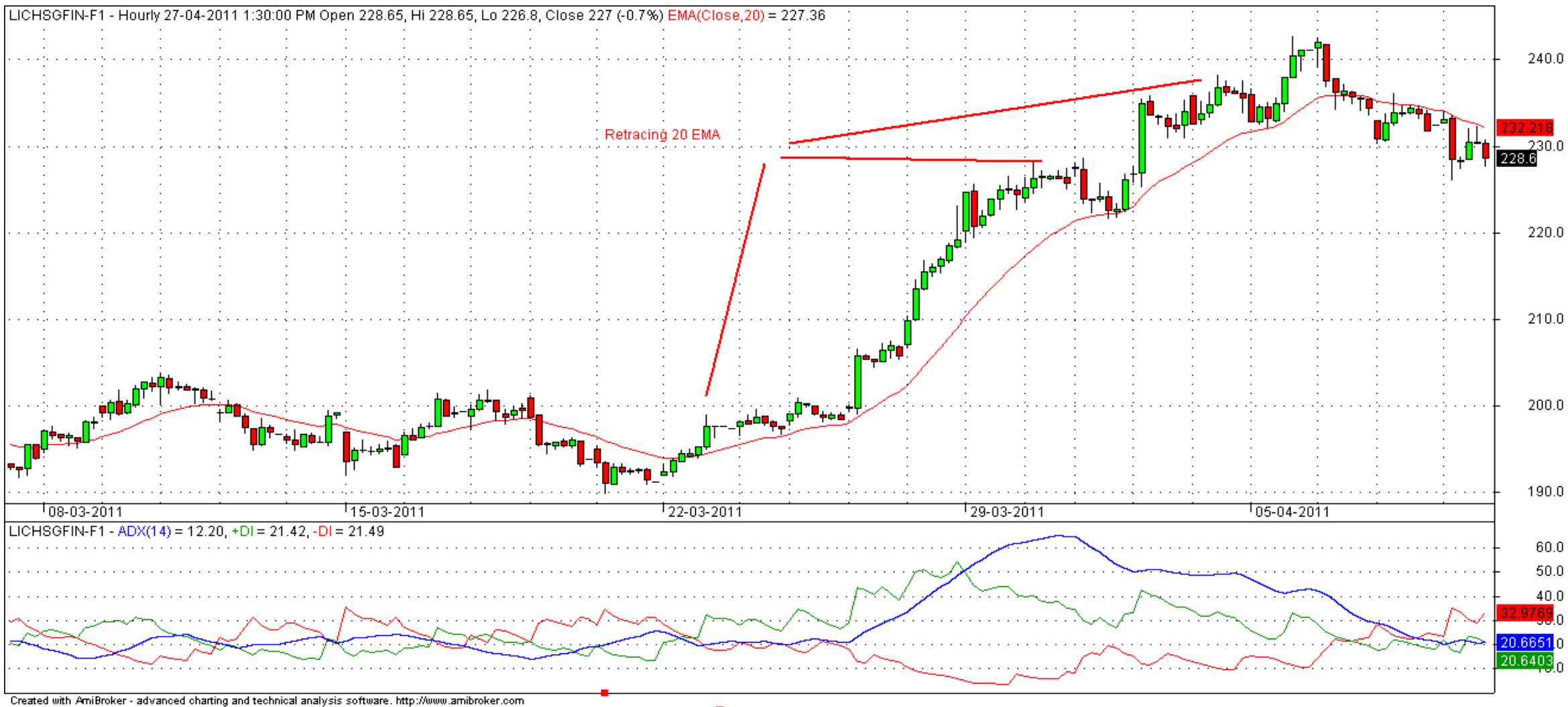
The Pullback, this strategy is taken from the book street smart, Linda call this strategy “The Holy grail” , this is a classic power trend strategy that finds price Retracing following a period of trend.

FOR BUYS (SELLS ARE REVERSED)

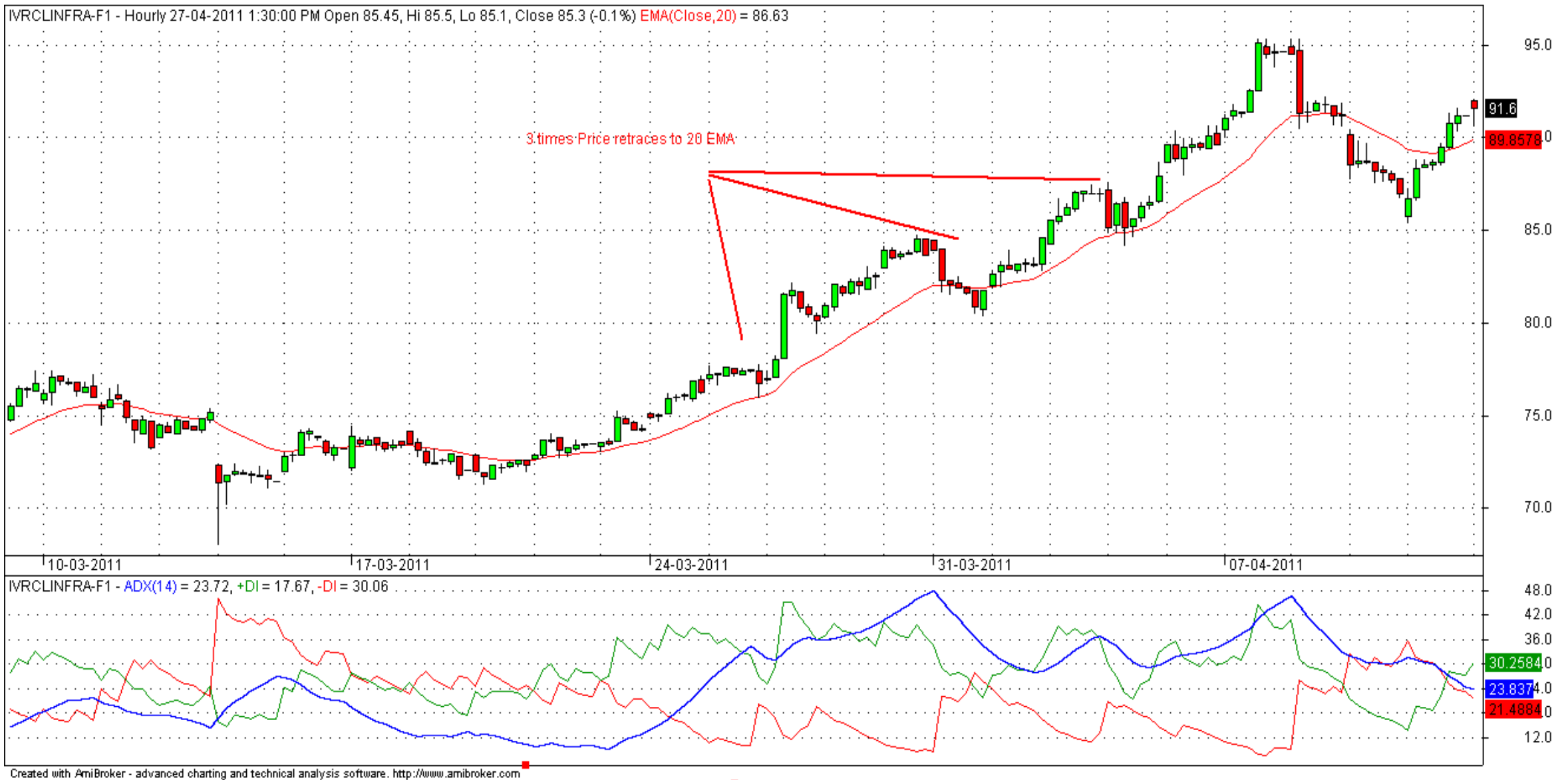
1. A 14-period ADX must initially be greater than 20 and rising. This will identify a strongly trending market.
2. Look for a retracement in price to the 20-period exponential moving average. Usually the price retracement will be accompanied by a turndown in the ADX.
3. When the price touches the 20-period exponential moving

average, put a buy stop above the high of the previous bar.

4. Once filled, enter a protective sell stop at the newly formed swing low. Trail the stop as profits accrue and look to exit at the most recent swing high. If you think the market may continue its move, you might exit part of the position at the most recent swing high and tighten stops on the balance.
5. If stopped out, re-enter this trade by placing a new buy stop at the original entry price.
6. After a successful trade, the ADX must once again turn up above 20 before another retracement to the moving average can be traded.



LICHSNG, 3 times Price came back to the EMA, remember in strong trend price generally retraces 2/3 times

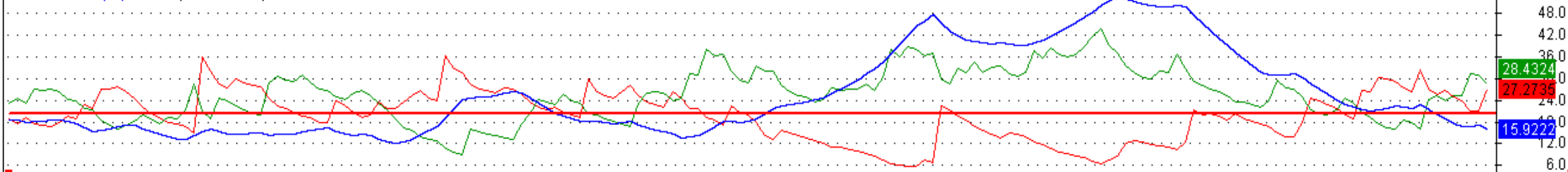


IVRCLINFRA, 3 times Price came back to the EMA, remember in strong trend price generally retraces 2/3 times

MNM-F1 - Hourly 27-04-2011 1:30:00 PM Open 770.5, Hi 771, Lo 766.3, Close 766.8 (-0.4%) EMA(Close,20) = 761.39



MNM-F1 - ADX(14) = 30.58, +DI = 30.83, -DI = 14.31



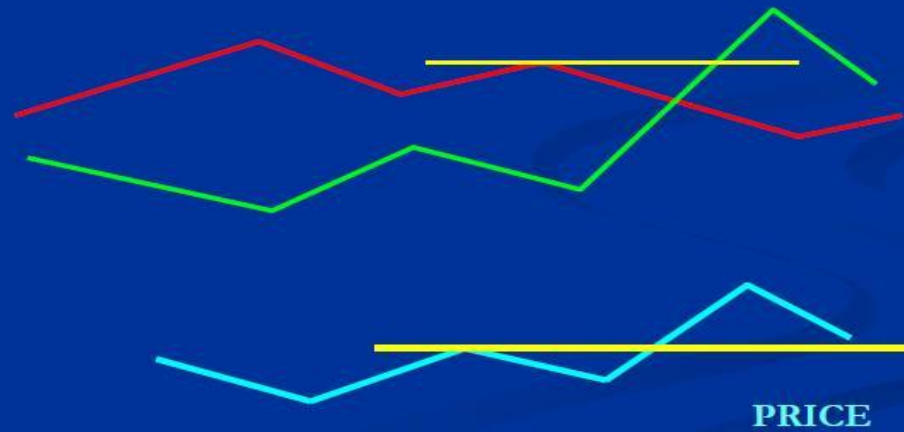
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MNM CHART

Reversal Pattern

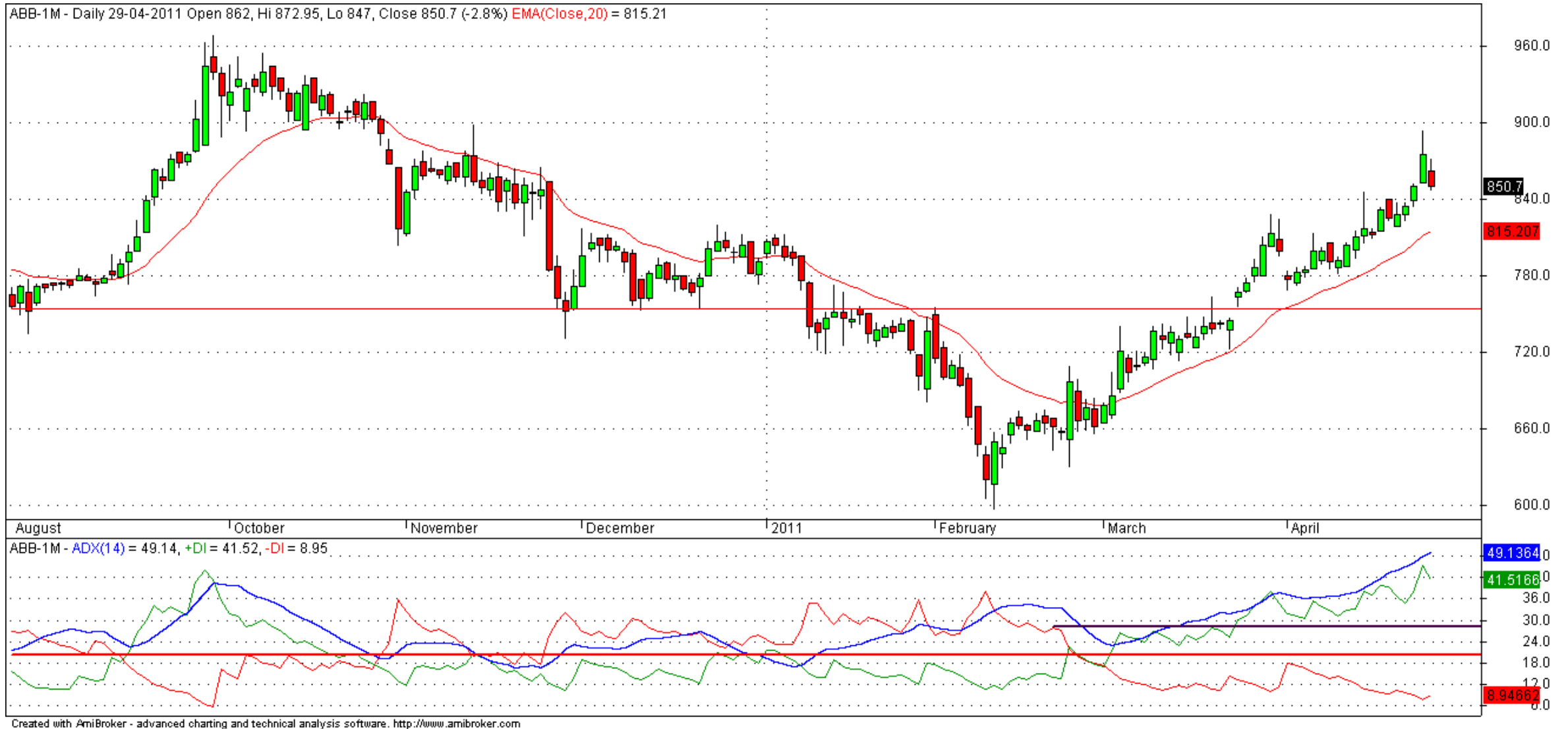
Point # 5

Crossover High (+DMI)
Change of Dominance





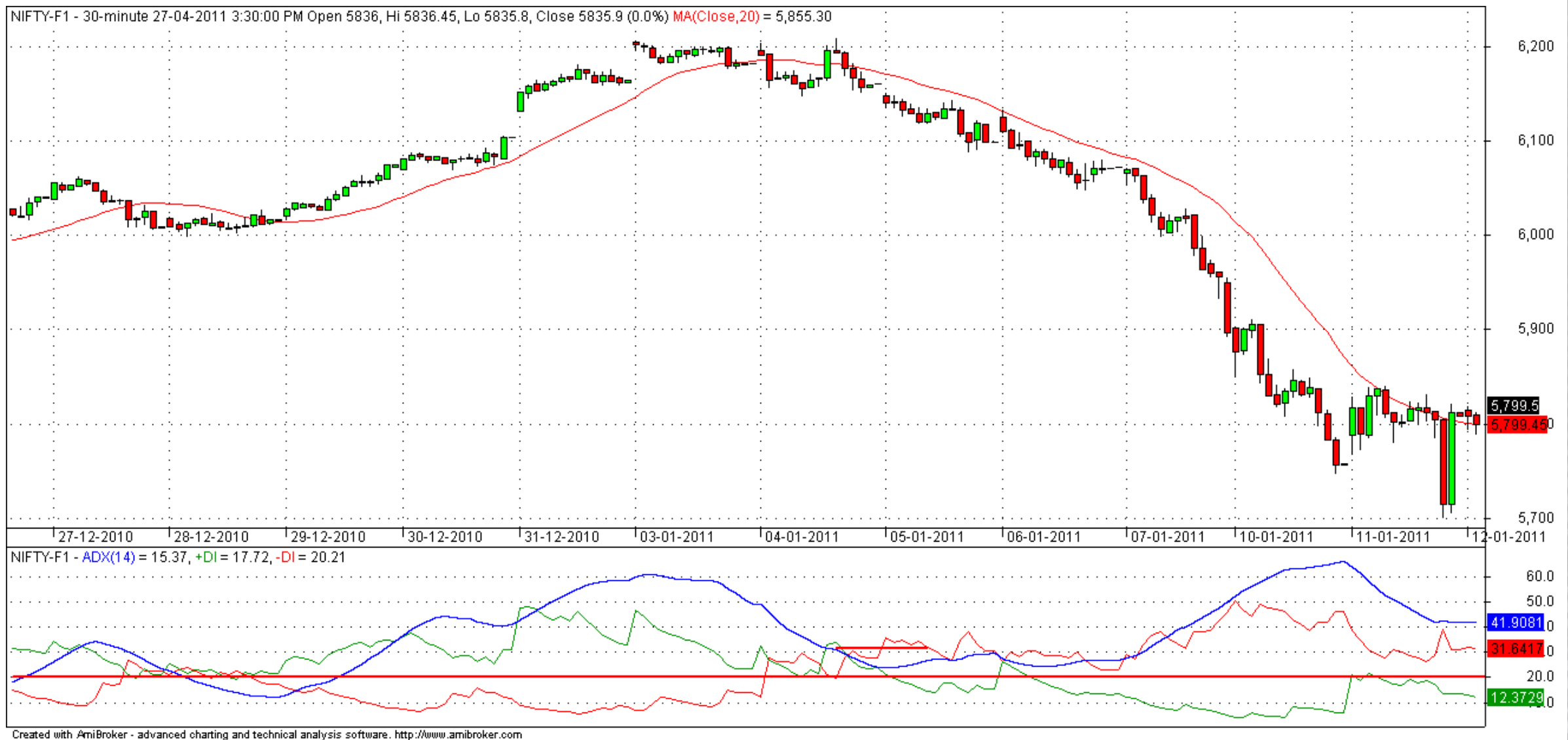
ALOKTEXT, on 04/04/2011, ADX remained above 20 and last high of -DMI was taken by +DMI, this is called Dominance shift



ABB, ADX never came below 20, the high of -DMI was taken over by +DMI, latter ADX confirmed the rally.



Nifty Chart, this is V reversal pattern, nifty went down and without any transition came back up, now see the red line drawn on DMI, it means dominance now has been taken over by bulls, when we see V sharp reversal we must see +DMI holds the dominance or -DMI by drawing trendline and see ADX must remain above 20



Nifty Chart, ADX Remained above 20 and bears took the dominance, the line drawn from the last high of +DMI is now taken over by -DMI, its called dominance shift, and later ADX also followed the trend.

Point # 6

Get Out of a Trend Trade With ADX Divergence

ADX Divergence Rule : When ADX is divergent with price, exit part or full position

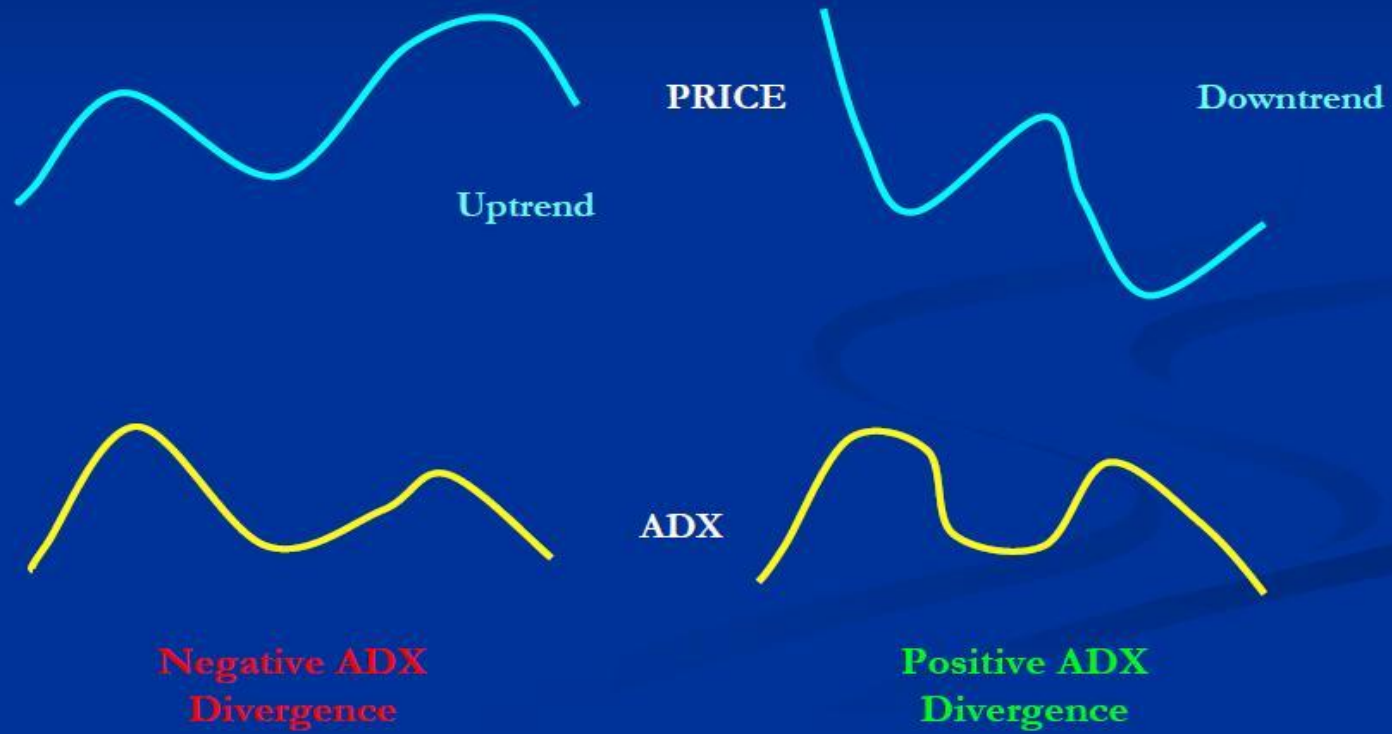
Most trading education focuses on making the perfect entry

Exits are harder and more important for profit consistency

Divergence = **D**ecision

Sell partial position (half) or full exit

ADX Divergences





Nifty Chart, ADX Remained above 20 and bears are in complete control, the first small line drawn from the last high of -DMI, +DMI couldn't able to break the high, dominance still remained with bears only, then we see a divergence in ADX value while ADX never came below 20, by next small line drawn on the high of -DMI was taken over by +DMI this is called dominance shift, now bulls are dominant and later ADX also confirmed the bull trend. It's a classic example of divergence with dominance shift.

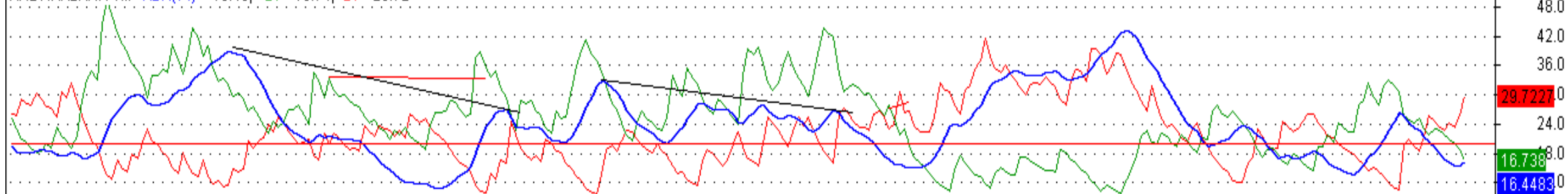
ANDHRABANK-1M - Daily 29-04-2011 Open 144, Hi 144.15, Lo 138.45, Close 139.55 (-2.1%) EMA(Close,20) = 145.83



Price and ADX are diverging, during the divergence
ADX stood above 20, making it a valid divergence

Divergence between price and ADX,
But ADX went below 20 before it made a divergence
making it a false divergence

ANDHRABANK-1M - ADX(14) = 16.45, +DI = 16.74, -DI = 29.72



POINTS TO REVIEW

- Point No. 1 **Low ADX Periods—No Trend Trading**
- Point No. 2 **20 Horse Power Minimum**
(When ADX is Below both DMI's try finding out patterns on the chart)
- Point No. 3 **Best Trades Begin From Low ADX Periods**
(A valid breakout requires a new high in price and a new high in +DMI (and +DMI > 20) , for Sell reverse.
- Point No. 4 **Pullbacks**
(ADX must initially be greater than 20 and rising, Look for a retracement in price to the 20-period exponential moving average. Usually the price retracement will be accompanied by a turndown in the ADX.)
- Point No. 5 **Reversals**
(Change of Dominance)
- Point No. 6 **Get Out of a Trend Trade With ADX Divergence**



THANK YOU
For Bearing with me so
long

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